2025_Hexaware_Actual_Test_9

Test Summary

- No. of Sections: 3
- No. of Questions: 92
- Total Duration: 130 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. One bell rings at an interval of 48 minutes and another at an interval of 92 minutes. If they both ring together at 11.00 am, the time when they will next ring together is

- 05.44 am
- 05.55 am
- 05.24 am
- 06.34 am

Q2. Find the least number which when divided by 5, 6, 7, 8 and 9 leaves reminders 3, 4, 5, 6 and 7 respectively.

- 2518
- 5036
- 1759
- 60

Q3. A shopkeeper gives a discount of 10% on the marked price of every shirt and when he sells 12 shirts he gives one shirt free. What is the approximate real rate of discount offered by the shopkeeper?

- 15%
- 17%
- 19%
- 12.5%

Q4. A man buys two articles. The ratio of the cost price of the articles is 3:2. He sells the first article at 20% profit and the second article at 10% loss. If he gains an overall Rs.40, find the cost price of the first article.

- 100
- 300
- 400
- 500

Q5. A shopkeeper in a town attracts a lot of customers to his shop by giving three successive discounts to the customers instead of the usual one time discount. On a festival season, he decided to give three discounts of 20%, 10% and another 5% successively. This successive discounts that he has given will be equal to a single discount of _____% ?

- 30%
- 31.6%
- 68.4%
- 35%

Q6. A dealer buys dry fruits at Rs 100, Rs 80 and Rs 60 per kilogram. He mixes them in the ratio 3 : 4 : 5 by weight and sells at a profit of 50%. At what price per kilogram does he sell the dry fruit?

- Rs 80
- Rs 100
- Rs 95
- None of these

Q7. Find the Compound Interest on Rs 30,000, if the rate of interest for the first year is 5%, for the second year is 10% and for the third year is 20%.

- 11580
- 11500
- 10500
- 10000

Q8.

Manivel invests two sum of money A and B at 10% p.a. and 20% p.a respectively at CI for 2 years. If the total interest on both the sum is Rs 5350 then find the sum invested in A if the total sum of A and B was Rs 20,000?

- Rs 5,000
- Rs 10,000
- Rs 12,000
- Rs 15,000

Q9. The average of the 100 results is 13, that of the first 40 is 12 and that of next 50 is 12½ . Find the average of the last 10 results.

- 18.5
- 19.5
- 15.5
- 16.5

Q10. Mohan and Ravi started a business by investing Rs.75000 and Rs.45,000 respectively. After 8 Months Keshav Joined by investing Rs. X. If at the end of the year, the profit earned by Keshav is equal to the average of profits earned by Mohan and Ravi together then find the value of X.

- Rs.180000
- Rs.170000
- Rs.165000
- Rs.150000

Q11. Mini started a business with investment amount of Rs. 54000 and after some months Mice joined her with investment money of Rs.72000. Mice invested his money for how many months if at the end of year profit is equally divided between them.

- 4
- 7
- 9
- None of these

Q12.

The income of A and B are in the ratio of 5: 3. The expenses of A, B and C are in the ratio 8: 5: 2. If C spends 2000 and B saves Rs. 700, then A saves

- Rs. 1000
- Rs. 2000
- Rs. 2500
- Rs. 1500

Q13. 11 men working for 18 hours a day can complete a piece of work in 25 days, in how many days can 18 men working for 5 hours a day complete the same piece of work?

- 50 days
- 48 days
- 52 day
- 55 days

Q14. If 3 men and 5 boys can do as much work in 17 days as 5 men and 3 boys can do in 15 days, then the ratio of rate of working of a man and a boy is

- 2 : 5
- 3 : 5
- 5 : 3
- 5 : 2

Q15. A fast train takes 3 hours less than a slow train for a journey of 600 km. If the speed of the slow train is 10 km/hr less than that of the fast train, then the speeds of two trains are

- 50, 40
- 60, 50
- 70, 80
- None of these

Q16. A train traveling at 48 kmph completely crosses another train having half its length and travelling in the opposite direction at 42 kmph, in 12 seconds. It also passes a railway platform in 45 seconds. The length of the platform is

400 m

450 m

560 m

600 m

Q17. Find the number of ways to distribute 15 chocolates among 4 persons if each gets at least 2 chocolates.

150

100

120

110

Q18. In how many ways can 5 boys and 7 girls be arranged in a row so that no two boys are together?

$7! \times {}^{7}P_{5}$

$7! \times {}^{8}P_{5}$

$8! \times {}^{7}P_{5}$

$7! \times {}^{8}P_{3}$

Q19. A number is chosen at random from 1 to 600 what is the probability of getting the numbers which is divisible by 20

1/20

1/15

2/17

1/6

Q20. In a given race, the odds in favor of horses H_1, H_2, H_3, and H_4 are 1:2, 1:3, 1:4, 1:5 respectively. Find the probability that one of them wins the race.

57/60

1/20

2/7

7/60

Q21. One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?

South

North

West

East

Q22. One morning Sujata started to walk towards the Sun. After covering some distance she turned to the right, then again to the right and after covering some distance she again turns to the right. Now in which direction is she facing?

North

South

West

East

Q23. **The question below consists of a question and two statements numbered (I) and (II) given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and answer.**

On which date in August was Kapil born?
I) Kapil's mother remembers that Kapil was born before the nineteenth but after the fifteenth.
II) Kapil's brother remembers that Kapil was born before the seventeenth but after the twelfth.

I alone is sufficient while II alone is not sufficient

II alone is sufficient while I alone is not sufficient

Either I or II is sufficient

Neither I nor II is sufficient

Both I and II are sufficient

Q24. **The question below consists of a question and two statements numbered (I) and (II) given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and answer.**

What is the code for 'or' in the code language?
I) 'nik sa te' means 'right or wrong', 'ro da nik' means 'he is right', and 'fe te ro' means 'that is wrong'.
II) 'pa nik la' means 'that right man', 'sa ne pa' means 'this or that', and 'ne ka re' means 'tell this there'.

- I alone is sufficient while II alone is not sufficient
- II alone is sufficient while I alone is not sufficient
- Either I or II is sufficient
- Neither I nor II is sufficient
- Both I and II are sufficient

Q25. In certain code language, '851' means 'good sweet fruit'; '783' means 'good red rose' '341' means 'rose and fruit'. Which of the following digits stands for 'sweet' in that language?

- 8
- 5
- 1
- 3

Q26. Using a particular code 'HILTON' is encrypted as 'IHTLNO'. Choose the right option for 'BILLION' coded in the same form.

- IBLLION
- IBOILLN
- IBLLOIN
- IBLOILN

Q27. Pointing to a photograph Lata says, "He is the son of the only son of my grandfather." How is the man in the photograph related to Lata ?

- Brother
- Son
- Father
- Nephew

Q28. Introducing Swathi, Rakesh said, “The only son of her father is my brother's mother's husband.” How is the man related to Swathi?

- Father
- Son
- Uncle
- Nephew

Q29. **Directions:**
What will come in place of question mark (?) in the following series?

9, 24, 69, 204, 609,?.

Q30. **Directions: What will come in place of question mark (?) in the following series?**

64, 88, 152, 162, 174, ?

Q31. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: I ≤ S, S < X, X = F, F ≤ K

Conclusion:
I. K > I
II. I < X

- if only conclusion 1 is true
- if only conclusion 2 is true
- if either conclusion 1 or conclusion 2 is true
- if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q32. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: S > P, P ≥ U, U > V, V ≤ N

Conclusion: I. N ≥ U II. S < N

if only conclusion 1 is true

if only conclusion 2 is true

if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q33. In the question below are given two or three statements followed by four conclusions numbered I, II, III and IV. You have to take the given statements to be true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follows from the statements, disregarding commonly known facts. Give answer.

Statements:
Some books are papers. All plates are records. Some records are books.

Conclusions:
I. Some plates are books.
II. Some records are papers.
III. Some papers are plates.
IV. Some books are records.

Only I follows

Only II and III follow

Only I and III follow

Only IV follows

Q34. In the question below are given two or three statements followed by four conclusions numbered I, II, III and IV. You have to take the given statements to be true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follows from the statements, disregarding commonly known facts. Give answer.

Statements:
All shirts are trousers. Some socks are shoes. All shoes are shirts.

Conclusions:
I. Some socks are shirts.
II. Some socks are trousers.
III. All shoes are trousers.
IV All shoes are socks.

Only I and II follow

Only I or II and III follows

Only II and IV follow

None of these

Q35. **In the following letter series, some of the letters are missing which are given in that order as one of the alternatives below it. Choose the correct alternative.**
b.... acbda.....bda... bd..... cb..... a.....

baadcd

dccadc

cbdcab

cdacbb

Q36. If the first and second letters of the word 'SUPERCILIOUS' are interchanged, also the third and fourth letters , fifth and sixth letters and so on , then which would be the seventh letter from the right?

R

L

P

C

Q37. **Arrange the sentences A, B, C, and D in a proper sequence to make a coherent paragraph.**

A. Such a national policy will surely divide and never unite the people.
B. In fact, it suits the purpose of the politicians; they can drag the people into submission by appealing to them in the name of religion.
C. In order to inculcate the unquestioning belief they condemn the other states, which do not follow their religion.
D. The emergence of the theocratic states where all types of crimes are committed in the name of religion, has revived the religion of the Middle Ages.

ABCD

DBCA

DBAC

CDAB

Q38. **A number of sentences are given below which, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.**

P) As Delhi heads for a caseload of half a lakh and 80 percent of ICU beds are filled, the health crisis is being intensified with nurses in the capital's private hospitals reportedly resigning or refusing to work.
Q) However, the plight of nurses, who have been working at the front in the battle against COVID-19, must be appreciated.
R) Their complaints are well-known, have been repeatedly articulated and remain largely ignored.
S) In some cases, nursing home owners are doubling up as janitors, while others are going to the police to complain that nurses are abandoning patients.

- PQSR
- PSQR
- SQRP
- SRQP

Q39. **Out of the four alternatives, choose the one which best expresses the meaning of the given word.**

Bereft

- Destitute
- Provide
- Supply
- Donate

Q40. **Identify the one which is opposite in meaning (antonym) to the question word and mark.**

crestfallen

- abashed
- cheerful
- broken
- treble

Q41. Identify the best way of writing the sentence in the context of the correct usage of standard written English. While doing so, you have to ensure that the message being conveyed remains the same in all cases.

One New York publisher has estimated that 50,000 to 60,000 people in the United States wanted an anthology that includes the complete works of William Shakespeare.

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Q42. Pick out the best alternative for the underlined part in the given sentences

If anyone dared to suggest that he or she was not against women but only against reservation, nobody was willing to listen

If anyone dared to suggest

If anyone was dared to suggest

If anyone dared to suggesting

If anyone dared suggest

Q43. Pick out the best alternative from the options given.

Whenever we went to the canteen, it was mandatory for each of us to pay our own bills.

to pay my own bill

paying our own bill

are paying our own bills

to pay his/ her own bill

Q44. **Change from direct speech to indirect speech**

Virat said, "I have passed the physical test."

He said that he had been passed the physical test.

Virat said that he has passed the physical test.

Virat said that he have passed the physical test.

Virat said that he had passed the physical test.

Q45. **Change from Direct speech to indirect speech.**

"You shouldn't swim in this sea," he said to me, "it's very dangerous."

He said that I shouldn't swim in this sea because it's very dangerous.

He said that you shouldn't swim in the sea if it was very dangerous.

He said that I shouldn't swim in that sea as it was very dangerous.

He said that you can't swim in this sea since it was very dangerous.

Q46. For the following questions, choose the option that fills up the blanks most suitably.

Your _____ and _____ behavior are stopping you from becoming a leader.

presumptuous, benign

callous, insubordinate

leadership, aggressive

laggard, leadership

Q47. Fill in the blank:
Listen to the story _____ answer the questions in complete sentences.

but

and

next

none of the above

Common Content:

Study the information given below and answer the questions based on it:

Nine persons namely J, K, L, M, N, O, P, Q and R live in a building which has 9 floor from 1 to 9, Lowermost floor is numbered 1 and top-most floor is number 9. They like different Actress viz, Deepika, Katrina, Sonam and Priyanka. Atleast two people like same actress not more than three people like same actress. L doesn't like Priyanka and doesn't live immediately above the one who likes Deepika. K lives on an odd numbered floor but not on the 1st floor. M lives on an even numbered floor. J who doesn't like Deepika lives three floors below the one who likes Sonam. R lives on a floor immediately above Q but not on 8th floor and 9th floor and they like different actress. The one who likes Katrina lives on 3rd floor. L lives four floors above N who likes same actress as M and none of them is either on 1st or top floor. The

persons on 8th and top floor are the only pair who likes the same actress (but not Sonam) and also live on adjacent floors and else no person sits adjacent to each other who likes the same actress. P who likes Priyanka lives on 2nd floor but doesn't live above the one who likes Sonam. M, who likes Katrina, lives immediately above R. The 7th floor is not occupied by Katrina.

Q48. Which actress is liked by three people?

- Deepika
- Sonam
- Katrina
- Priyanka

Q49. Which of the following statement is correct?

- M lives on 6th floor and likes Deepika
- M lives on 4th floor and likes Katrina
- Q doesn't like Sonam
- None of the given options

Q50. Who lives two floors below R?

- The one who likes Katrina
- The one who likes Sonam
- The one who likes Priyanka
- L

Q51. Who among the following likes Deepika?

- K
- O
- L
- Both K and O

Common Content:

Read the passage and answer the question that follows

The Indian middle class consists of so many strata that it defies categorization under a single term class, which would imply a considerable degree of homogeneity. Yet two paradoxical features characterize its conduct fairly uniformly; extensive practice and intensive abhorrence of corruption.

In several recent surveys of popular perceptions of corruption, politicians of course invariably and understandably top the list, closely followed by bureaucrats, policemen, lawyers, businessmen, and others. The quintessential middle class. If teachers do not figure high on this priority list, it is not for lack of trying, but for lack of opportunities. Over the years, the sense of shock over acts of corruption in the middle class has witnessed a steady decline, as its ambitions for a better material life have soared but the resources for meeting such ambitions have not kept pace.

What is fascinating, however, is the intense yearning of this class for a clean corruption-less politics and society, a yearning that has again and again surfaced with any figure public or obscure, focus on his mission of eradicating corruption. Even the repeated failure of this promise on virtually every man's part has not subjected it to the law of diminishing returns.

Q52. Teachers are not high on the list of corruption because they do not have

- Courage
- Opportunities
- Support
- Ambition

Q53. Who figure on top of the list of corruption?

- businessmen
- lawyers
- politicians
- policemen

Q54. The Indian Middle class is

- defiant
- mysterious
- homogenous
- stratified

Q55. The intense Middle Class intensely yearns for

- better material resources
- extensive practice of corruption
- clean honest society
- law of increasing returns

Common Content:

The Ordinance to raise the foreign capital cap in insurance from 26% to 49% is not the best way to do it, as it ____(1)____the parliament _____(2)_____. Presumably, however, US President Barack Obama's visit explains the government's tearing hurry. New Delhi wants to show that it is resolute in ____(3)____ reform. However, it would be naive to expect that foreign investors will rush to raise their ___(4)____ in India joint ventures and infuse more funds in the sector that needs lots of capital go grow. Investors want certainty in law. Foreign partners will ____(5)____ and watch to see how things pan out as an Ordinance is a stopgap arrangement. Eventually, Parliament must approve the law.

Q56. Choose the appropriate word for the blank numbered (5)

- halt
- stay
- stop
- wait

Q57. Choose the appropriate word for the blank numbered (4)

- stakes
- goals
- roles
- risks

Q58. Choose the appropriate word for the blank numbered (1)

- encourages
- circumvents
- hoodwinks
- evades

Q59. Choose the appropriate word for the blank numbered (2)

- procedure
- inaction
- process
- growth

Q60. Choose the appropriate word for the blank numbered (3)

- presuming

aggressive

pushing

hasty

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode:

```
1  Integer a,b,c
2  Set a=7,b=7,c=9
3  if ((a^b^c)>(b^a))
4     c=b&b
5  End if
6  Print a+b+c
```

23

21

24

27

Q2. What will be the output of the following pseudocode?

```
1  Start
2     Integer p, q, r
3     Set p = 0, q = 8, r = 4
4
5     For each r from 5 to 8
6        Set q = 5 + 12 + p
7        If (r - q - p) < (3 - r) Then
8           Set p = q + p
9        Else
10          Set q = p
11       End If
12    End For
13
14    Print p + q
15 End
```

291

290

391

390

Q3. What will be the output of the following pseudocode?

```
1  Integer p , q , r , s
2  Set p = 15, q = 0, r = 60, s = 0
3  while ((r MOD p ) EQUALS 0)
4      s = ( s+2) * 2
5      q = q + p
6      while (s < 50 )
7        Print s
8        q = q - 1
9         s = s + q
10     end while
11     p = p -3
12 end while
13 print q
```

4 28 60 23

19 27 54 63 23

4 18 31 43 23

4 18 31 43 26

Q4. What will be the output of the following pseudocode?

```
1  Integer j, m
2  Set m=2
3  Integer a[4] = {2,1,1,2}
4  m=m^a[2]
5  if (a[1]>1)
6     m=m^1
7  Else
8     m=m^2
9  End if
10 Print m
```

8

-1

1

12

Q5. What will be the output of the following pseudocode if p=5, q=10, r=15?

```
1  void fun(Integer p, Integer q, Integer r)
2  p= p>>1
3  q= q>>1
4  r= r>>1
5  p= p>>1
6  q= q>>1
7  r= r>>1
8  Print p+q+r
9  End function fun()
```

7

5

6

10

Q6. What will be the output of the following pseudocode?

```
Integer c, d, e
Set d = 22
for(each c from 2 to 6)
  d= d+c
  e= c+d
end for
e=d/5
Print c,d,e
```

6 41 7

6 42 40

7 42 8

7 41 40

Q7. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=6,b=6,c=7
for (each c from 3 to 5)
    a=(8+4)+a
End for
c=(6+5)+a
for (each c from 5 to 7)
    b=(3&7)+b
    a=a^b
End for
Print a+b
```

47

45

56

53

Q8. What will be the output of the following pseudocode?

```
Integer p, q, r
set p=1, q=1, r=1
if (0 | | 1 | | 1 )
    p= p & 1
else
    r = r & 1
```

```
7  end if
8  print p + q + r
```

- 18
- -1
- 10
- 3

Q9. What will be the output of the following pseudocode when i=5?

```
1  Integer fun ( Integer i)
2    if((i/2) < (i MOD 2))
3     return i + 1
4    else
5     return fun(fun (i-6))
6   end if
7  End function fun()
```

- -6
- -4
- 12
- 13

Q10. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=9,b=6,c=6
3  for (each c from 5 to 9)
4     if((c+a)>(a-c))
5        a=(b+c)+a
6        b=7&c
7     End if
8     a=a^c
9  End for
10 Print a+b
```

- 47
- 75
- 60
- 58

Q11. What will be the output of the following pseudocode?

```
1  Integer a,b,c
```

```
2  Set a=7,b=5,c=5
3  for (each c from 4 to 5)
4     if (c>b)
5        Jump out of the loop
6     End if
7     b=a+b
8     b=(b+a)+c
9  End for
10 Print a+b
```

- 49
- 50
- 61
- 44

Q12. What will be the output of the following pseudocode when a=2 and b=7?

```
1  Integer funn(Integer a, Integer b)
2     if (b>a && b>6)
3        return funn(b,a)+funn(b,1)
4     End if
5     return 1
```

- 16
- 3
- 2
- -2

Q13. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=2,b=7,c=4
3  a=(c&b)^a
4  if ((c+5)<(5+c)
5     c=(c+c)+b
6     b=(10+3)^b
7     c=(b^a)^a
8  End if
9  Print a+b+c
```

- 20
- 16
- 17

24

Q14. What will be the output of the following pseudocode for x = 108 and y = 3?

```
1  Integer fun (Integer x, Integer y)
2     if (x > 1)
3        fun (x/y, y+3)
4        Print y
5     enf if
6  end function fun ()
```

12 9 6

9 6 3

0 6 36 108

6 12

Q15. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=5, q=8, r=6
3  p=p+q
4  p=(6+5)+p
5  for(each r from 2 to 6)
6   p=r+q
7   if((p+r)>(q-p) OR r<q)
8     q=(r&p)^p
9     Continue
10  End if
11 End for
12 Print p+q
```

40

22

44

33

Q16. Which of the following is not used as secondary storage?

Semiconductor memory

Magnetic disks

Magnetic drums

Magnetic tapes

Q17. Which of the following printers uses light beam and electrostatically sensitive black powder?

- dot matrix printer
- daisy wheel printer
- chain printer
- laser printer

Q18. A table consists of

- Fields and Columns
- Rows and Columns
- Rows and Cells
- None of the mentioned options

Q19. Supercomputers are mainly useful for

- Mathematical intensive scientific applications
- Data-retrieval operations
- Input-output intensive processing
- All of the mentioned options

Q20. Which of the following is true about primary storage?

- It is a part of the CPU
- It allows very fast access to data
- It is relatively more expensive
- All of the mentioned options

Q21. Which device can draw continuous lines

- daisy wheel

- plotter
- chain printer
- impact printer

Q22. In comparison to the internal (main) memory, tape or disk memory is

- slower and more expensive
- slower and less expensive
- faster and more expensive
- faster and less expensive

Q23. Which technique is preferable for transferring a large amount of data to and from a memory in a short time?

- DMA
- Interrupt driven I/O
- Programmed I/O
- None of the mentioned options

Q24. People who gain unauthorized access to computers for the fun of it, but do not intentionally do damage, are

- Employees
- Hackers
- Crackers
- Members of organized crime

Q25. Small files that are deposited on user's hard derive when they visit a website are called

- Cookies
- Codes
- Profiles

Trackers

Q26. What is wired Equivalent Privacy (WEP)?

Security algorithm for Ethernet

Security algorithm for wireless networks

Security algorithm for USB communication

None of the mentioned options

Q27. What is the access point (AP) in wireless LAN?

Device that allows wireless device to connect to a wired network

Wireless devices itself

Both Device that allows wireless device to connect to a wired network and Wireless devices itself

None of the mentioned options

Q28. Which of the following is an application layer service?

Remote login

Mail service

File transfer and access

All of the mentioned options

Q29. Which one of the following event is not possible in wireless LAN?

Collision detection

Acknowledgement of data frames

Multi-mode data transmission

None of the mentioned options

Q30. The attributes of message security are

Integrity

Authentication

Non-repudiation

All of the mentioned options

Section 3 - Coding

Section Summary

- No. of Questions: 2
- Duration: 40 min

Additional Instructions:
None

Q1. **Problem Statement**

Given a string **s** consisting of words and spaces, return the length of the last word in the string.

Example 1:
Input: s = "Hello World"
Output:
5
World
Explanation: The last word is "World" with length 5.

Example 2:
Input: s = "luffy is still joyboy"
Output:
6
joyboy
Explanation: The last word is "joyboy" with length 6.

Input Format

The input consists of a string **s.**

Output Format

The first line of output prints the length of the last word in the string.
The second line prints the last word in the string.

Refer to the sample output for the formatting specifications.

Constraints

1 ≤ s.length ≤ 100
s consists of only English letters and spaces ' '.
There will be at least one word in s.

Sample Input

```
Hello World
```

Sample Output

```
5
World
```

Sample Input

```
luffy is still joyboy
```

Sample Output

```
6
joyboy
```

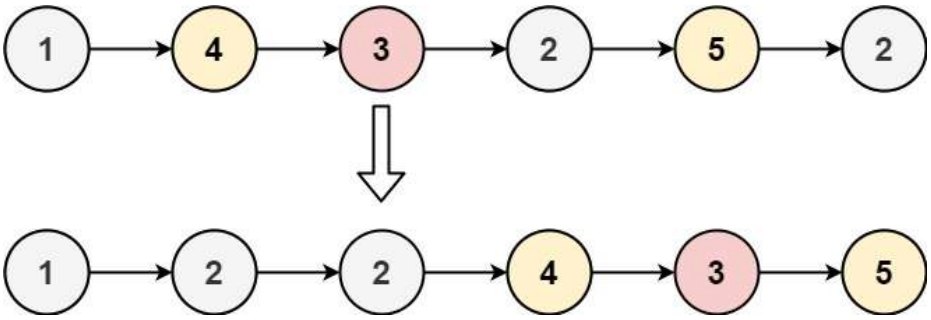
Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Problem Statement**

Given the head of a linked list and a value x, partition it such that all nodes less than x come before nodes greater than or equal to x.

You should preserve the original relative order of the nodes in each of the two partitions.

Example 1



Input:
6
1 4 3 2 5 2
3

Output:
1->2->2->4->3->5

Example 2

Input:
2
2 1
2

Output:
1->2

Input Format

The first line of input consists of an integer **N** representing the number of nodes in the linked list.
The second line of input consists of **N** space separated integers representing the elements of the nodes in the linked list.
The third line of input consists of an integer **X** representing the partition value.

Output Format

The output prints the partitioned linked list separated by an arrow ->.

Refer to the sample output for the formatting specifications.

Constraints

In this scenario, the given test cases will fall under the following constraints:
2 ≤ N ≤ 8
1 ≤ Elements of the node ≤ 200
1 ≤ X ≤ 200

Sample Input

```
6
1 4 3 2 5 2
3
```

Sample Output

```
1->2->2->4->3->5
```

Sample Input

Sample Output

```
2
2 1
2
```

```
1->2
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Answer Key & Solution

Section 1 - Aptitude

Q1

05.24 am

Solution

Take the LCM of 48 and 92 = 1104 minutes = 18 hrs 24 min after 11.00 am i.e., 5:24 am.

Q2

2518

Solution

LCM of 5, 6, 7, 8, 9 = 2520.

Now the difference in the divisor and remainder for all 5 numbers are constant = 2.

Hence, Answer = 2520 –2=2518.

Q3

17%

Solution

MP of one shirt = x

SP of shirt = 12x * 90/100 = 10.8x

Real discount = ((12 + 1)x - 10.8x)/(12 + 1)x * 100

= 17%

Q4

300

Solution

Let the CP of two articles be 3x and 2x

3x*20/100 - 2x*10/100 = 40

x =100

Cp of first article = 300

Q5

31.6%

Solution

Consider from the values 100

20% from 100 will result in 80

10% from 80 will result in 72

5% from 72 will result in 68.4

So, the difference in amount is 31.6 and hence the total discount % will also be 31.6%

Q6

None of these

Solution

Cost price of(3 + 4 +5) =12 kg of fruits

=Rs(300 + 320 + 300) =Rs 920

SP at a profit of 50% =Rs 1380

SP of fruits per kg =1380/12 =Rs. 115

Q7

11580

Solution

Amount = (Principal + Compound interest) = P(1 + R1/100)(1 + R2/100)(1 + R3/100)

=30000(1 +5/100)(1 +10/100)(1 +20/100)

=41580

Interest = 41580-30000 = 11580

Q8

Rs 15,000

Solution

At 10% CI in 2 years=21 %

At 20% Ci in 2 years =44%

and 5350 is 107/4% of 20000, by using allegation

A B

21 44

107/4

3 1

A=3/4x20000= Rs 15000

Q9

19.5

Solution

100*13 = 40*12 + 50*12.5 + 10*x => x = 19.5

Q10

Rs.180000

Solution

Keshav invested Rs.X for a period of 4 months.Ratio of profit sharing = Mohan : Ravi : Keshav 75000*12 : 45000*12 : X * 4As per the question,(75000*12) + (45000*12) / 2 = 4XHence, X = 180000 Rs.

Q11

9

Solution

Let Mice joins after x months.

Then Mice's investments was for (12-x) months

Since the profits was divided equally.

54000×12=72000(12-x)

X=3

Mice's joined after 3 months or we can also say that he invested his money for 9 months.

Q12

Rs. 1500

Solution

No Solution

Q13

55 days

Solution

We know that work = M × D × H

Here,

M = no. of men

D = no of days

H = hours

So,

$M_1 \times D_1 \times H_1 = M_2 \times D_2 \times H_2$

$11 \times 25 \times 18 = 18 \times D_2 \times 5$

$\therefore D_2 = 55$ days

Q14

5 : 3

Solution

By comparing the work, (3M + 5B) × 17 = (5M + 3B) × 15

=> 51M + 85B = 75M + 45B

=> 40B = 24M =>

M/B = 40/24 = 5/3

M:B = 5 : 3.

Q15

50, 40

Solution

Let the speed of fast train = x km./hr.

Then the speed of slow train = (x - 10) km./hr.

Time taken for journey of 600 km by fast train =(600/x) hours.

Time taken for journey of 600 km by slow train = (600/(x-10)) hours.

Since the fast train takes 3 hours less than the slow train.

(600/x-10) - (600/x) =3

x = 50 or - 40.

Rejection x = - 40, as the speed of fast train cannot be negative.

Speed of fast train = x = 50 km./hr

Speed of slow train = x - 10 = 40 km./hr.

Q16

400 m

Solution

Let the length of the first train be x metres.

Then, the length of the second train is x/2 m

Relative speed = (48 + 42) kmph =(90 x 5/18)m/sec = 25 m/sec.

Therefore [x + (x/2)]/25 = 12

= > x = 200.

Therefore Length of first train = 200 m.

Let the length of platform be y metres.

Speed of the first train =(48 x 5/18)m/sec. = 40/3 m/sec

Therefore (200 + y) x 3/40 = 45

=> 600 + 3y = 1800

=> y = 400 m.

Q17

120

Solution

Since each student gets atleast 2 chocolates

4 x 2 = 8

15 - 8 = 7

$^{n+r-1}C_{r-1}$

$= 7 + 4 - 1\ C_{4-1}$

$= 10C_3 = 120$

Q18

$7! \times {}^8P_5$

Solution

No Solution

Q19

1/20

Solution

numbers divisible by 20 between 1 to 600 = 600/20 = 30

probability of numbers divisible by 20 bet 1 to 600 = 30C1/600C1 = 30/600 = 1/20

Q20

57/60

Solution

$P(H_1) = 1/3,$

$P(H_2) = 1/4,$

$P(H_3) = 1/5,$

$P(H_4) = 1/6$

$P(H_1 \cup H_2 \cup H_3 \cup H_4) = P(H_1) + P(H_2) + P(H_3) + P(H_4) = 1/3 + 1/4 + 1/5 + 1/6 = 57/60.$

Q21

North

Solution



Q22 North

Solution

Towards sun in the morning means towards east. Hence finally Sujata will face towards North.

Q23 Both I and II are sufficient

Solution

From I, we conclude that Kapil was born on any one of the dates among 16th, 17th and 18th.

From II, we conclude that Kapil was born on any one of the dates among 13th, 14th, 15th and 16th.

Thus, from both I and II, we conclude that Kapil was born on 16th August.

Q24 Either I or II is sufficient

Solution

I. In 'right or wrong' and 'he is right', the common word is 'right' and the common code word is 'nik'. So 'nik' means 'right'. In 'right or wrong' and 'that is wrong', the common word is 'wrong' and the common code word is 'te'. So, 'te' means 'wrong'.

Thus, in 'right or wrong', 'sa' is the code for 'or'. II. In 'that right man' and 'this or that', the common word is 'that' and the common code word is 'pa'. So, 'pa' means 'that'. In 'this or that' and 'tell this there', the common word is 'this' and the common code word is 'ne'. So, 'ne' means 'this'. Thus, in 'this or that', 'sa' is the code for 'or'.

Q25 5

Solution

No Solution

Q26 IBLLOIN

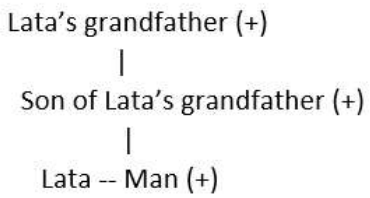
Solution

Firstly letters are grouped together in a pair of two such as BI, LL, IO, N. The position of the respective letters is changed in all the pairs which make them IB, LL, OI, N. Hence, the code for BILLION becomes IBLLOIN.

Q27 Brother

Solution

Male(+), Female(-), Couple(=), Siblings(--), Next Generation (|)



The man in the photograph is the son of the only son of Lata's grandfather i.e., the man is the son of Lata's father. Hence, the man is the brother of Lata.

Q28 Nephew

Solution

Rakesh brother's mother's husband is Rakesh's Father.

Swathi's father is Rakesh's Father.

So, Swathi and Rakesh are siblings.

So, the son of Swathi is Rakesh's Nephew

Q29 1824

Solution

Series Pattern Given Series

9 9

9 + 15 = 24 24

24 + 15 × 3 = 69 69

69 + 15 × 3 × 3 = 204 204

204 + 15 × 3 × 3 × 3 = 609 609

609 + 15 × 3 × 3 × 3 × 3 = 1824 1824

Q30 202

Solution

64 + (6 × 4) = 88

88 + (8 × 8) = 152

152 + (1 × 5 × 2) = 162

162 + (1 × 6 × 2) = 174

Q31 if both conclusion 1 and conclusion 2 are true

Solution

Combined Inequalities:

I ≤ S < X = F ≤ K I ≤ S < X = F ≤ K∴ I < K.

Hence, Conclusion I is true.

I ≤ S < X = F ≤ K∴ I < X. H

Hence, Conclusion II is true.

Q32 if neither conclusion 1 nor conclusion 2 is true

Solution

Combined inequalities:

S > P ≥ U > V ≤N S > P ≥ U > V ≤ N,

No relationship cannot be established between N and U.

Hence, conclusion I is not true.

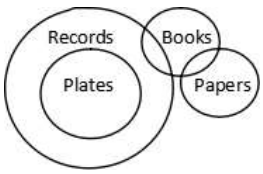
S > P ≥ U > V ≤ N,

No relationship cannot be established between N and S.

Hence, conclusion II is not true.

Q33 Only IV follows

Solution



Conclusions:-

I False

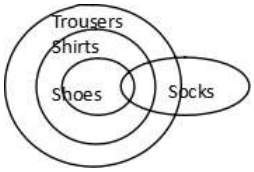
II False

III False

IV True

Q34 None of these

Solution



Conclusions:

I True

II True

III True

IV False

None of the answer options are right.

Q35 dccadc

Solution

The series is: bdac/bdac/bdac/bdac.

Q36 R

Solution

S U P E R C I L I O U S

U S E P C R L I O I S U - after rearrangement

7th letter from the right is R.

Q37 DBCA

Solution

CA is a pair. The 'policy' in A refers to the condemnation of the states that do not follow a particular religion. The opening sentence of the paragraph is D and this is immediately followed by B. So, the order of sentences is DBCA.

Q38 PSQR

Solution

In the sequence **PSQR**, the paragraph begins by highlighting the escalating health crisis in Delhi due to a shortage of nurses in private hospitals (P). It then depicts the immediate consequences of nurses resigning, showcasing nursing home owners' desperate measures and complaints (S). Following this, it acknowledges the vital role of nurses in battling COVID-19 and the need to appreciate their plight (Q). Finally, it emphasizes the persisting issues faced by nurses, whose grievances have been consistently ignored despite being well-known (R).

The correct answer is Option (2)

Q39 Destitute

Solution

Bereft => to be sad and lonely because of the death of some near one.

Destitute => to be alone and depressed due to poverty-stricken

Provide, Supply and Donate => to give to someone in need.

Q40 cheerful

Solution

'Crestfallen' means dejected. 'Abashed' is humiliated. 'Treble' means the pitch range of highest female voice. Hence, 'cheerful' is the right choice.

Q41 One New York publisher has estimated that 50,000 to 60,000 people in the United States want an anthology that includes the complete works of William Shakespeare.

Solution

3 is the only choice that is grammatically correct. Choices 1 and 4 use the verbs incorrectly. Choice 2 uses a instead of an before anthology.

Q42 If anyone dared suggest

Solution

2 and 3 are incorrect. The 'to' is unnecessary in the sentence, so the answer is 4.

Q43 to pay his/ her own bill

Solution

The verb references singular noun each of us.

So sentence should be his/her own bill.

Q44 Virat said that he had passed the physical test.

Solution

When the direct speech becomes indirect speech, **I have** becomes **he had.**

Q45 He said that I shouldn't swim in that sea as it was very dangerous.

Solution

The sentence is talking about the obligation to not swim in the sea considering the danger.

You should be changed to I

this to be changed to that.

And simple past tense verb should be used.

Q46 callous, insubordinate

Solution

Ans. (2). Obviously two negative words should fill in-the blanks. Only option (2) is qualified to be the apt answer. All other options contain a negative and a positive word. So, options (1), (3) and (4) can be sidelined.

Q47 and

Solution

and-used to connect two clauses when the second refers to something that happens after the first.

Q48. Katrina

Solution

No Solution

Q49. None of the given options

Solution

Floor	Person	Actress
9	K	Deepika
8	O	Deepika
7	L	Sonam
6	M	Katrina
5	R	Priyanka
4	Q	Sonam
3	N	Katrina
2	P	Priyanka
1	J	Katrina

Q50. The one who likes Sonam

Solution

o Solution

Q51. Both K and O

Solution

Floor	Person	Actress
9	K	Deepika
8	O	Deepika
7	L	Sonam
6	M	Katrina
5	R	Priyanka
4	Q	Sonam
3	N	Katrina
2	P	Priyanka
1	J	Katrina

Q52. Opportunities

Solution

From the following line of the passage "teachers do not figure high on this priority list, it is not for lack of trying, but for lack of opportunities" the answer is opportunities.

Q53. politicians

Solution

From the line, "the several recent surveys of popular perceptions of corruptions, politicians of course invariably and understandably top the list..." it is clear that politicians top the list.

Q54. homogenous

Solution

From the line, "The Indian middle class consist of so many strata that it defies categorization under a single term class, which would imply a considerable degree of homogeneity." It is clear that the answer is homogeneous

Q55. clean honest society

Solution

From the line, "What is fascinating, however, is the intense yearning of this class for a clean corruption less politics and society" it is clear that the society is yearning for clean honest society.

Q56. wait

Solution

o Solution

Q57. stakes

Solution

The word "**stakes**" is appropriate in this context as it refers to the share or interest that individuals or entities have in a venture or business.

Q58. circumvents

Solution

The word "**circumvents**" is suitable in this context as it means to avoid or bypass through clever and indirect methods. Therefore, the sentence is expressing the idea that the Ordinance is taking an indirect route to achieve the increase in the foreign capital cap without going through the regular parliamentary process.

Q59. process

Solution

The word "**process**" fits well in this context, as it refers to a series of actions or steps taken to achieve a particular result.

Q60. pushing

Solution

The word "**pushing**" is appropriate in this context, as it conveys the idea of actively advocating or promoting a reform agenda. Therefore, the sentence is expressing that New Delhi aims to demonstrate its commitment to reform by actively pushing or promoting it.

Section 2 - Pseudocode and Computer Fundamentals

Q1 21

Solution

If block executes and update c=7&7 = 7.Finally 7+7+7=21.

Q2 391

Solution

No Solution

Q3

4 18 31 43 23

Solution

No Solution

Q4

1

Solution

No Solution

Q5

6

Solution

No Solution

Q6

7 42 8

Solution

When c=2, updated value of d=24,e=26

When c=3, updated value of d=27,e=30

When c=4, updated value of d=31,e=35

When c=5, updated value of d=36,e=41

When c=6, updated value of d=42,e=48 and increment c=7 then terminates loop.

So, the output is 7,42,8.

Q7

47

Solution

Inside first loop:

When c=3, a=18

When c=4, a=30

When c=5, a=42

Inside second for loop:

When c=5, update value is a=35,b=9

When c=6, update value is a=47,b=12

When c=7, update value is a=32,b=15

So 32+15 = 47.

Q8

3

Solution

No Solution

Q9

-4

Solution

In fun(5) , 5/2<5%2 (false) so fun(fun(-1)) called.

In fun(-1) , -1/2<-1%2 (true) so return 0.

Then in fun(0) , 0/2 < 0%2(false) so fun(fun(-6)) called.

In fun(-6) , -6/2 < -6%2(true) so return -5.

In fun(-5) , -5/2 < -5%2(true) so return -4.

Finally -4 gets printed.

Q10

58

Solution

When c=5, If block gets executed and a=(6+5+9)= 20, b=7&5=5,a=20^5=17.Same process repeated till c=9. So a=57,b=1. Finally 57+1=58.

Q11

49

Solution

When c=4, if block gets failed and update b=7+5 = 12, b=12+7+4= 23.

Same process repeated for next iteration and update b=42.So 7+42=49.

Q12

2

Solution

In funn(2,7) => return funn(7,2)+funn(7,1)

In funn(7,2) => return 1.

In funn(7,1) => return 1.

In funn(2,7) => 1+1 =>2.

Q13

17

Solution

a=(4&7)^2 = 6. If block condition gets failed, So 6+7+4 = 17.

Q14

9 6 3

Solution

No Solution

Q15

22

Solution

Initially p=5,q=8,r=6, When r =2 p and q value updated to 10 and 8.

When r =3 p and q value updated to 11 and 8.

When r =4 p and q value updated to 12 and 8.

When r =5 p and q value updated to 13 and 8.

When r =6 p and q value updated to 14 and 8 and terminates loop. Finally 14+8=22 gets printed.

Q16

Semiconductor memory

Solution

No Solution

Q17

laser printer

Solution

No Solution

Q18

Rows and Columns

Solution

No Solution

Q19

Mathematical intensive scientific applications

Solution

No Solution

Q20

All of the mentioned options

Solution

Primary storage is the component of the computer that holds data. It is a part of the CPU, programs, and instructions that are currently in use. Primary storage is located on the motherboard. As a result, data can be read from and written to primary storage extremely quickly.

Q21

plotter

Solution

No Solution

Q22

slower and less expensive

Solution

No Solution

Q23

DMA

Solution

No Solution

Q24

Hackers

Solution

No Solution

Q25

Cookies

Solution

No Solution

Q26

Security algorithm for wireless networks

Solution

No Solution

Q27

Device that allows wireless device to connect to a wired network

Solution

No Solution

Q28

All of the mentioned options

Solution

No Solution

Q29

Collision detection

Solution

No Solution

Q30

All of the mentioned options

Solution

No Solution

Section 3 - Coding

Q1

Test Case

Input	Output
Programming is fun	3 fun

Weightage - 10

Input	Output
Example HI	2 HI

Weightage - 10

Input	Output
The quick brown fox jumps	5 jumps

Weightage - 15

Input	Output
Spaces at the end	3 end

Weightage - 15

Input	Output
These test cases cover a range of scenarios	9 scenarios

Weightage - 25

Input	Output
Medium Level Test Cases Hard Level Test Cases	5 Cases

Weightage - 25

Sample Input

```
Hello World
```

Sample Output

```
5
World
```

Sample Input

```
luffy is still joyboy
```

Sample Output

```
6
joyboy
```

Solution

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <stdbool.h>

struct Result {
    int length;
    char* lastWord;
};

struct Result lengthOfLastWord(char* s) {
    int length = 0;
    char* lastWord = (char*)malloc(strlen(s) + 1);
    lastWord[0] = '\0';
    bool counting = false;

    for (int i = strlen(s) - 1; i >= 0; i--) {
        if (s[i] != ' ') {
            counting = true;
            length++;
            char temp[2] = {s[i], '\0'};
            char* newWord = (char*)malloc(strlen(lastWord) + 2);
            strcpy(newWord, temp);
            strcat(newWord, lastWord);
            free(lastWord);
            lastWord = newWord;
        } else if (counting) {
            break;
        }
    }

    struct Result res;
    res.length = length;
    res.lastWord = lastWord;
    return res;
}

int main() {
    char input[1000];
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';  // Remove the newline character

    struct Result result = lengthOfLastWord(input);
    printf("%d\n", result.length);
    printf("%s\n", result.lastWord);

    free(result.lastWord);
```

```
    return 0;
}
```

```
#include <iostream>
using namespace std;

class Solution {
public:
    pair<int, string> lengthOfLastWord(string s) {
        int length = 0;
        string lastWord;
        bool counting = false;

        for (int i = s.length() - 1; i >= 0; i--) {
            if (s[i] != ' ') {
                counting = true;
                length++;
                lastWord = s[i] + lastWord;
            }
            else if (counting) {
                break;
            }
        }

        return make_pair(length, lastWord);
    }
};

int main() {
    Solution obj;
    string input;
    getline(cin, input);
    pair<int, string> result = obj.lengthOfLastWord(input);
    cout << result.first << endl;
    cout << result.second << endl;
    return 0;
}
```

```
import java.util.Scanner;

class Solution {
    public Result lengthOfLastWord(String s) {
        int length = 0;
        String lastWord = "";
        boolean counting = false;

        for (int i = s.length() - 1; i >= 0; i--) {
            if (s.charAt(i) != ' ') {
                counting = true;
                length++;
                lastWord = s.charAt(i) + lastWord;
            } else if (counting) {
                break;
            }
        }

        return new Result(length, lastWord);
    }
}

class Result {
    public final int length;
```

```
    public final String lastWord;

    public Result(int length, String lastWord) {
        this.length = length;
        this.lastWord = lastWord;
    }
}

public class Main {
    public static void main(String[] args) {
        Solution obj = new Solution();
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        Result result = obj.lengthOfLastWord(input);
        System.out.println(result.length);
        System.out.println(result.lastWord);
        scanner.close();
    }
}
```

Q2

Test Case

Input

```
5
3 1 2 4 5
3
```

Output

```
1->2->3->4->5
```

Weightage - 10

Input

```
4
2 2 2 2
3
```

Output

```
2->2->2->2
```

Weightage - 10

Input

```
7
5 6 3 8 2 9 4
7
```

Output

```
5->6->3->2->4->8->9
```

Weightage - 15

Input

```
6
10 5 12 8 15 7
9
```

Output

```
5->8->7->10->12->15
```

Weightage - 15

Input

```
8
9 7 4 12 18 2 10 15
11
```

Output

```
9->7->4->2->10->12->18->15
```

Weightage - 25

Input	Output
8 90 70 40 120 140 200 100 150 200	90->70->40->120->140->100->150->200

Weightage - 25

Sample Input	Sample Output
6 1 4 3 2 5 2 3	1->2->2->4->3->5

Sample Input	Sample Output
2 2 1 2	1->2

Solution

```
#include <iostream>

struct ListNode {
    int val;
    ListNode* next;
    ListNode(int x) : val(x), next(nullptr) {}
};

ListNode* partition(ListNode* head, int x) {
    ListNode* lessHead = new ListNode(0);  // Dummy node for nodes less than x
    ListNode* lessTail = lessHead;
    ListNode* greaterHead = new ListNode(0);  // Dummy node for nodes greater than or equal to x
    ListNode* greaterTail = greaterHead;

    ListNode* curr = head;

    while (curr) {
        if (curr->val < x) {
            lessTail->next = curr;
            lessTail = lessTail->next;
        } else {
            greaterTail->next = curr;
            greaterTail = greaterTail->next;
        }
        curr = curr->next;
    }

    lessTail->next = greaterHead->next;  // Connect the two partitions
    greaterTail->next = nullptr;  // Set the end of the greater partition

    return lessHead->next;
}

int main() {
    // Create the linked list
    int n, val, x;
    std::cin >> n;  // Number of nodes
```

```
    std::cin >> val;  // Value of the first node
    ListNode* head = new ListNode(val);
    ListNode* curr = head;

    for (int i = 1; i < n; ++i) {
        std::cin >> val;
        curr->next = new ListNode(val);
        curr = curr->next;
    }

    std::cin >> x;  // Partition value

    // Call the partition function
    ListNode* newHead = partition(head, x);

    // Print the partitioned linked list
    curr = newHead;
    while (curr) {
        std::cout << curr->val;
        if (curr->next) {
            std::cout << "->";
        }
        curr = curr->next;
    }

    return 0;
}
```